Supplementary Material for "Executable Conformance Profiles for Differentiable Numerical Components"

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1. Calibration details

Table S1 reports the matched optimizer-level calibration used in the main text. The two methods used the same noisy observations, parameter bounds, initialization, and five random seeds. Work is reported in the native unit of each optimizer and should not be interpreted as an optimizer ranking.

Table 1: Matched two-parameter matrix-action calibration. Values are mean ± standard deviation over five seeds unless stated otherwise.

Method	Parameter err.	Final loss	Work	Time (s)
Adam/autodiff	0.002060 ± 0.000700	4.116×10^{-6}	140 updates	1.368
L-BFGS-B/classical	0.002076 ± 0.000741	4.116×10^{-6}	27 evaluations	0.02835

2. Backend-specific numerical evidence

The entries in Table S2 retain the measured quantity and the tested range used to support the cross-primitive profile. These values have different backend-specific meanings and are not pooled into a single score.

3. Batch, device, and precision measurements

All GPU timings use synchronized execution on the RTX 5070 Laptop GPU. Memory is incremental CUDA memory measured by the experimental harness. The results characterize this implementation and hardware configuration.

Table 2: Representative value, gradient, OOD, and long-horizon evidence.

Backend	Test condition	Observed error
MLSL	80-digit series reference	value 1.32×10^{-12}; α gradient 2.22×10^{-9}; β gradient 5.00×10^{-10}
Matrix action	$t = 0.02$–12, five seeds	long-horizon maximum absolute error 4.27×10^{-15}
Linear RK4	OOD step size 0.41–0.80	maximum absolute error 1.67×10^{-3}
Linear RK4	$t = 5$, step sizes 0.05 and 0.20	1.48×10^{-9} and 4.23×10^{-7}
Logistic RK4	OOD step size 0.26–0.50	maximum absolute error 7.52×10^{-6}

Table 3: Float64 batch profile over 50 timed iterations after warmup.

Backend and batch	Latency (ms)	Throughput (samples/s)	Memory (bytes)
Matrix, 1	0.510062	1,960.55	8,525,824
Matrix, 64	0.945798	67,667.73	8,549,376
Matrix, 256	0.942034	271,752.40	8,635,392
Matrix, 1024	0.928594	1,102,742.43	8,979,456
Linear RK4, 1	0.333228	3,000.95	8,525,312
Linear RK4, 64	0.340564	187,923.56	8,530,432
Linear RK4, 256	0.357048	716,990.43	8,561,152
Linear RK4, 1024	0.355290	2,882,152.61	8,684,032
Logistic RK4, 1	0.430600	2,322.34	8,525,312
Logistic RK4, 64	0.438784	145,857.64	8,525,824
Logistic RK4, 256	0.444710	575,656.04	8,542,720
Logistic RK4, 1024	0.454500	2,253,025.31	8,610,304

Table 4: Dtype-specific precision and resource measurements.

Workload	Dtype	Value error	Gradient error	Latency (ms)	Memory (bytes)
Matrix action	float32	4.3229×10^{-7}	4.2235×10^{-8}	1.03765	210,432
Matrix action	float64	1.7764×10^{-15}	1.0645×10^{-10}	1.06621	418,304
Periodic heat	float32	2.8978×10^{-7}	1.8410×10^{-7}	0.298242	1,312,256
Periodic heat	float64	1.6376×10^{-15}	1.8773×10^{-16}	0.306160	2,624,000

4. Requirement traceability matrix

Table S5 connects the project requirements used by the reference validator to the standards concerns discussed in the main paper. The cited standards motivate the concern; the CQ identifiers and decision rules are defined by this study and are not clauses of those standards.

Table 5: Traceability from standards concerns to executable component-level evidence.

ID	Standards concern	Schema field	Test action	Artifact and decision
CQ-01	Identity and version control	schema, component, implementation	Validate required identifiers and versions	Validated record; missing or unknown identifier is nonconformant
CQ-02	Planned test scope and completion	operating domain, coverage	Freeze scope; check case count and required anchors	Coverage diagnostics; unfrozen or insufficient scope is nonconformant
CQ-03	Functional accuracy measurement	evidence.value, tolerances, units	Compare with an independent value reference under mixed gates	Error record and value-gate decision
CQ-04	Derivative reliability	evidence.gradient, perturbation scale	Directional finite-difference sweep and finite-gradient check	Gradient error, stable-step region, and gate decision
CQ-05	Reliability and repeatability	batch, repeatability	Heterogeneous-batch isolation and repeated evaluation	Shape, isolation, and repeatability evidence
CQ-06	Compatibility and execution policy	requested and observed execution	Compare API/CLI, dtype, device, and fallback behavior	Execution record; mismatch is nonconformant
CQ-07	Measurement traceability	units, provenance, reference	Check completeness and consistency of units and lineage	Provenance and unit fields bound to evidence
CQ-08	Stress-test planning	OOD, long horizon	Execute frozen boundary and horizon probes	Error trajectory and first failed regime
CQ-09	Application evidence	calibration, composition	Run matched calibration and at least two compositions	Application-profile evidence and task decision
CQ-10	Portable exchange and change control	canonical JSON, digest, migration provenance	Key-order, API–CLI, and legacy-migration tests	Stable digest; migrated records require requalification